

Article

A co-optimization algorithm utilizing Particle Swarm Optimization for Linguistic Time Series

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Abstract: Linguistic time series forecasting model (LTS-FM), which has been recently proposed, uses linguistic words of linguistic variable domain generated by hedge algebras (HAs) to describe historical numeric time series data. Then, the LTS-FM is established by utilizing real numeric semantics of words which induced by the fuzziness parameter values (FPVs) of HAs. In the existing LTS-FMs, just the FPVs of HAs are optimized whilst the used word set is still chosen by human experts. This paper proposes a co-optimization method of selecting the optimal used word set which best describes numeric time series data in parallel with choosing best FPVs of HAs to improve the accuracy of LTS-FM by utilizing particle swarm optimization (PSO). In this co-optimization method, the outer loop optimizes the FPVs of HAs whilst the inner loop optimizes the used word set. The experimental results on three datasets of the “enrollments of the University of Alabama” (EUA), the “killed in car road accidents in Belgium” (CAB), and the “spot gold in Turkey” (SGT) show that our proposed forecasting model outperforms the existing forecasting models in terms of forecasted accuracy.

Keywords: Linguistic Time Series; Hedge Algebras; Linguistic Logical Relationship; Particle Swarm Optimization; Forecasting Model.

MSC:

Citation: To be added by editorial staff during production.

Academic Editor: Firstname Last-name

Received: date

Revised: date

Accepted: date

Published: date



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1. Introduction

The human ability to predict future events and phenomena has attracted the scientific community for many years with a bunch of proposed forecasting methodologies based on observed historical data. In particular, the forecasting method based on time series analysis has been investigated by numerous researchers with various models such as ARMA, ARIMA, and so on. In time series forecasting models, future values can be forecasted based on only past data. The characteristics of the time series data such as trends, seasonality, stability, outliers, etc. have been considered to establish forecasting models for future values.

The fuzzy time series proposed by Q. Song & B. S. Chissom [1–3] is a forecasting method of time series analysis that combines the principles of the fuzzy set theory proposed by L. A. Zadeh [4] with traditional time series techniques. Fuzzy time series is particularly useful in handling the uncertainty, vagueness, and imprecision of data, which is

often encountered in real-world applications. The approach allows for modeling the uncertainty and vagueness in the data by representing the relationships between linguistic variables as fuzzy sets rather than crisp sets. In fuzzy time series, the forecasted values are not simply single point estimates, but rather a range of values that reflect the uncertainty in the data. This provides a more realistic and robust representation of future trends in the time series, especially when dealing with non-linear and complex data. Fuzzy time series has been applied in a variety of domains, including finance, economics, environmental science, and engineering, where traditional time series methods may not be adequate. Recently, many proposed models apply optimization algorithms such as genetic algorithm [5, 6], particle swarm optimization (PSO) [7, 8], etc. to optimize the forecasting model parameter values to improve the forecasting accuracy.

People often use natural language as a tool to communicate effectively with each other. They also store and process recorded information through linguistic variables with their values. Linguistic values are also used by humans to forecast events occurring in nature and society so that they can better prepare for future planning. Beside fuzzy set theory for handling linguistic variables, hedge algebras (HAs) proposed by N. C. Ho and W. Wechler [9, 10] exploits the order-based semantics structure of word domains of linguistic variables which provides a mathematical formalism for generating computational semantics of words that can be applied to solve the real-life application problems in various domains such as image processing [11], fuzzy control [12], data mining [13–17], etc. HAs is the qualitative model, therefore, it needs to be quantified by measurable quantities based on qualitative semantics. The words of linguistic variables convey their qualitative semantics in relationship with the other words in the whole variable domain. Because the linguistic words of a variable are interdependent, they only disclose their full qualitative semantics in the comparative context of the whole variable domain. The fuzziness measure of linguistic words [18] is the crucial concept of fuzzy information and plays an important role in hedge algebras quantification. The quantitative or numerical semantics of a word induced by the fuzziness measures of words is the semantically quantifying mapping (SQM) value that is the basis for generating the computational semantics for solving real-world application problems. Recently, hedge algebras have been applied to solve the time series forecasting problem in such a way that the historical numeric time series data is transformed into linguistic one by using the real numeric semantics of words defined based on their corresponding SQM values. Then, the LTS-FMs are established [19–22].

In the LTS-FMs, instead of partitioning the historical data into intervals, a datum of historical data is assigned a linguistic word based on the nearest real numeric semantic. By doing so for all historical data, the numeric time series is transformed into LTS. The linguistic words used to describe historical data are used to define the logical relationship between the data of the current year and the next year. Then, the linguistic logical relationships (LLRs) and linguistic logical relationship groups (LLRGs) are established. The fuzzy forecasted values are induced and finally based on them, the corresponding crisp forecasted values are computed. We can see that the SQM values of words determined by the FPVs of HAs are crucial in establishing the LTS-FMs. The FPVs of HAs are automatically optimized by PSO in [23]. However, the word set used to describe numeric time series data is chosen by human experts, so the selected words may not reflect the nature of the historical data. To overcome this drawback, this paper presents a method of selecting an optimal word set which best describe numeric time series data in parallel with optimizing the FPVs of HAs by applying a co-optimization algorithm of particle swarm optimization (Co-PSO). In Co-PSO, the outer loop optimizes the FPVs of HAs whilst the inner loop optimizes the used word set. The experimental results on three datasets of the “enrollments of the University of Alabama” (EUA), the “killed in car road accidents in Belgium” (CAB), and the “spot gold in Turkey” (SGT) show that our proposed forecasting model has better forecasting accuracy than the existing ones.

The remainder of this paper is organized as follows. Section 2 briefly restates the theory of hedge algebras, some concepts related to LTS-FM and PSO. The proposed forecast-ing model called COLTS will be introduced in Section 3. In Section 4, some experiments with three datasets and discussions about the forecasted results will be addressed. Finally, we summarize this work and suggest some lines of research for future work.

2. Background

2.1. A brief introduction of hedge algebras

Each linguistic domain of variable $\mathcal{X}$, denoted by $\text{Dom}(\mathcal{X})$, consists of a word set that can be generated from two generator words, e.g., “cold” and “hot” by acting linguistic hedges on them. For example, with two hedges of “very” and “rather”, the words gener-ated by acting two those hedges on two generator words of “cold” and “hot” can be “very cold”, “rather cold”, “very hot”, “very very hot”, and so on. We can observe that they are linearly ordered and are comparable, e.g., “very cold” $\leq$ “cold” $\leq$ “rather cold” $\leq$ “hot” $\leq$ “very hot” $\leq$ “very very hot”.

By the above intuitive observation, Ho et al. introduced hedge algebras (HAs) in 1990 [9, 10], mathematic structure, which can directly manipulate the word domain of $\mathcal{X}$. A HA of $\mathcal{X}$, denoted by $\mathcal{AX}$ is an order-based structure $\mathcal{AX} = (X, G, C, H, \leq)$, where $X \subseteq \text{Dom}(\mathcal{X})$ is a word set of $\mathcal{X}$; $G = \{c, c^+\}$ is a set of two generators, where $c \leq c^+$; $C = \{0, W, 1\}$ is a set of three constants satisfying the order relation $0 \leq c \leq W \leq c^+ \leq 1$, where 0, 1, and W are the least, greatest, and neutral constants, respectively; A set of hedges of $H = H^- \cup H^+$, where H^- and H^+ are the negative and positive hedges, respectively; $\leq$ is an operator which indi-cates the order relation between the inherent word semantics of $\mathcal{X}$.

String representation can be used to represent the words in X , so a word $x \in X$ is either c or ωx , where $c \in \{c, c^+\}$, $\omega = h_n \dots h_1$, $h_i \in H$, $i = 1, \dots, n$. $H(x)$ denotes all words generated from x , so we have $H(x) = \{\omega x, \omega \in H\}$. If all hedges in H are linearly ordered, all words in X are also linearly ordered. In this case, we have linear HAs. Hereafter are some main properties of the linear HAs:

- The signs of the negative generator c and the positive generator word c^+ are $\text{sign}(c) = -1$ and $\text{sign}(c^+) = +1$, respectively.
- Every $h \in H^+$ makes the semantic of c^+ increased and has $\text{sign}(h) = +1$, whereas, every $h \in H^-$ makes the semantic of c decreased and $\text{sign}(h) = -1$.
- If the hedge h strengthens the semantic of the hedge k , the relative sign between h and k is $\text{sign}(h, k) = +1$. On other side, if the hedge h weakens the semantic of the hedge k , $\text{sign}(h, k) = -1$. Thus, the sign of a word $x = h_n h_{n-1} \dots h_2 h_1 c$ is specified as follows: $\text{sign}(x) = \text{sign}(h_n, h_{n-1}) \times \dots \times \text{sign}(h_2, h_1) \times \text{sign}(h_1) \times \text{sign}(c)$.

Based on the word sign, we have $\text{sign}(hx) = +1 \rightarrow x \leq hx$, $\text{sign}(hx) = -1 \rightarrow hx \leq x$.

Based on the syntactical semantics of words generated by an HAs, the words in $H(x)$, $x \in X$, induced from x have been changed their semantics by the hedges in H , but they still convey the original semantics of x . Therefore, $H(x)$ can be considered as the fuzziness of x and the diameter of $H(x)$ is considered as the *fuzziness measure* of x , denoted by $fm(x)$.

Assume that $\mathcal{AX}$ is a linear HAs. A function $fm: X \rightarrow [0, 1]$ is *fuzziness measure* of the words in X provided that the following properties are satisfied [13]:

(F1): $fm(c) + fm(c^+) = 1$ and $\sum_{h \in H} fm(hu) = fm(u)$, for $\forall u \in X$;

(F2): $fm(x) = 0$ for all $H(x) = x$, especially, $fm(0) = fm(W) = fm(1) = 0$;

(F3): $\forall x, y \in X, \forall h \in H$, the proportion $\frac{fm(hx)}{fm(x)} = \frac{fm(hy)}{fm(y)}$ is called the fuzziness meas-ure of the hedge h , denoted by $\mu(h)$.

From the properties of (F1) and (F3), $fm(x)$, where $x = h_n \dots h_1 c$ and $c \in \{c, c^+\}$, is com-puted as $fm(x) = \mu(h_n) \dots \mu(h_1) fm(c)$, where $\sum_{h \in H} \mu(h) = 1$. For a given word in X , the fuzz-iness measure of it can be computed when the values of $fm(c)$ and $\mu(h_i) \in H$ are specified.

Semantically quantifying mapping (SQM) [13] of $\mathcal{AX}$ is a mapping $v: X \rightarrow [0, 1]$ pro-vided that the following conditions are satisfied:

(SQM1): it preserves the order based structure of X , i.e., $x \leq y \rightarrow v(x) \leq v(y), \forall x \in X$;

(SQM2): It is one-to-one mapping and $v(x)$ is dense in $[0, 1]$.

Let fm be a fuzziness measure on X , $\sum_{i=-q}^{-1} \mu(h_i) = \alpha$, $\sum_{i=1}^p \mu(h_i) = \beta$, $\alpha, \beta > 0$ and $\alpha + \beta = 1$. $v(x)$ is computed recursively based on fm as follows:

1) $v(W) = \theta = fm(c^-)$, $v(c^-) = \theta - \alpha fm(c^-) = \beta fm(c^-)$, $v(c^+) = \theta + \alpha fm(c^+)$;

2) $v(h_j x) = v(x) + sign(h_j x)(\sum_{i=sign(j)}^j fm(h_i x) - \omega(h_j x)fm(h_j x))$, where

$j \in [-q \wedge p] = \{j: -q \leq j \leq p \text{ \& } j \neq 0\}$ and

$$\omega(h_j x) = \frac{1}{2} [1 + sign(h_j x)sign(h_p h_j x)(\beta - \alpha)] \in \{\alpha, \beta\}.$$

The SQM values of words are the basis for computed the real numerical semantics of words, and then a time series is transformed into LTS.

2.2. Linguistic time series forecasting model

Based on the theory of hedge algebras, Hieu et al. in [19] introduced the concept of LTS and its application to the enrollment forecasting problem. In which, the quantitative (numerical) semantics, known as SQM values, of linguistic words are directly used to establish the LTS-FM. Specifically, the numerical semantics of words are linearly transformed to the real numerical semantic domain of the universe of discourse (UD) of the linguistic variable. Thus, each datum of a time series, whether recorded in numerical or linguistic values, can be natural associated with the corresponding real numerical semantics of the used linguistic words of the LTS. This is also the distinguished and outstanding properties of LTS compared with fuzzy time series.

In order to have a theoretical basis in proposing our forecasting model, we briefly present in this subsection some concepts of LTS proposed by Hieu et al. [19].

Definition 1. [19] (LTS) Let X be a set of natural linguistic words of a variable X defined on the UD to describe its numeric values. Then, any chronological series $L(t)$, $t = 0, 1, 2, \dots$, where $L(t)$ is a finite subset of X , is called a linguistic time series.

Definition 2. [19] (LLR) Suppose X_i and X_j are the linguistic words representing the data at the time t and $t+1$, respectively. Then, there exists a relationship between X_i and X_j called a linguistic logical relationship (LLR) and denoted by

$$X_i \rightarrow X_j.$$

Definition 3. [19] (LLRG) Assume that there are LLRs such as

$$X_i \rightarrow X_{j1},$$

$$X_i \rightarrow X_{j2},$$

$$\dots$$

$$X_i \rightarrow X_{jn}.$$

Then, they can be grouped into a linguistic logical relationship group (LLRG) and denoted by

$$X_i \rightarrow X_{j1}, X_{j2}, \dots, X_{jn}.$$

In [19], Hieu et al. also presented a forecasting model called LTS-FM. The procedure of LTS-FM is depicted as in Figure 1.

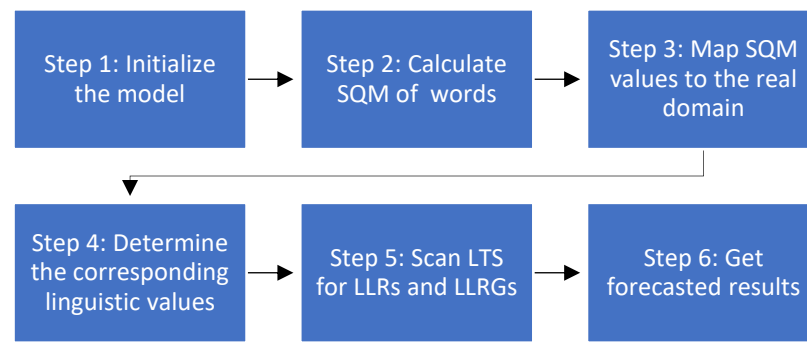


Figure 1. The procedure of LTS-FM.

In Figure 1, the LTS-FM procedure includes six steps which can be briefly described as follows (for more detail, please see [19, 22]):

Step 1. Determine the UD of the linguistic variable, the syntactical semantics and the FPVs of the associated HAs, choose the used linguistic words to describe the designated time series.

Step 2. Calculate the quantitative semantics of the used words.

Step 3. Map the quantitative semantics of the used words to the real domain of the UD to get the real numerical semantics.

Step 4. Transform the designated time series into LTS. For each specified datum, the semantics of it is specified based on the nearest real semantic.

Step 5. Establish the LLRs of words, then group them into the LLRGs.

Step 6. Forecast based on LLRGs, then compute the crisp forecasted values.

The evaluation measures of the forecasting models are the mean square error (MSE), the root mean square error (RMSE), and the mean absolute percentage error (MAPE) which are calculated as follows:

$$MSE = \frac{1}{n} \sum_{i=1}^n (F_i - R_i)^2, \quad (1)$$

$$RMSE = \sqrt{MSE}, \quad (2)$$

$$MAPE = \frac{1}{n} \sum_{i=1}^n \left| \frac{F_i - R_i}{F_i} \right| \times 100\%. \quad (3)$$

where n denotes the number of forecasted data, F_i and R_i are forecasted and real data at the time i , respectively.

2.2. Standard PSO

In 1995, Kennedy and Eberhart introduced an optimization method, so-called Particle swarm optimization (PSO). It is a swarm intelligence-based optimization method that has been applied to solve a lot of real-world problems. It mimics the way the birds fly to find food sources. The birds in the swarm will follow the leader who is the nearest to the food source. Suppose that a swarm $S = \{x_1, x_2, \dots, x_N\}$ with N particles which each its particle position X_i^t at generation t is computed as follows:

$$X_i^{t+1} = X_i^t + V_i^{t+1}, \quad (4)$$

where V_i^{t+1} is x_i 's velocity at generation $t + 1$ computed as follows:

$$V_i^{t+1} = \omega \times V_i^t + c_1 \times r_1 \times (P_i^t - X_i^t) + c_2 \times r_2 \times (P_g^t - X_i^t), \quad (5)$$

where P_g^t and P_i^t are the best global and local solutions found so far, respectively; r_1 and r_2 are two random numbers uniformly distributed in $[0, 1]$; c_1 , c_2 , and ω are self-cognitive coefficient, social cognitive coefficient, and inertia weight, respectively. The standard PSO is described in brief as follows:

Step 1: Initialize a swarm S with two random vectors, the position vector X and velocity vector V . Initialize the number of cycle constant N and the cycle variable t .

Step 2: Calculate each particle's objective value $f(X_i^t)$.

Step 3: Check each particle's objective. If current position X_i^t is better than the personal best P_i^t then update P_i^t with X_i^t .

Step 4: Check if there exists the best particle position in the current cycle whose objective value is better than the objective value of the global best P_g^t then update P_g^t with the best particle position.

Step 5: Update each particle's velocity V_i^t by Eq. (4) and move to its new position X_i^t by Eq. (5).

Step 6: Terminate if variable t reaches the maximum number of cycles, otherwise let $t=t+1$ and goto Step 2.

3. Improve the LTS-FM by co-optimization of PSO

In this subsection, a hybrid LTS-FM integrated with particle swarm co-optimization (COLTS) is presented. In the LTS-FMs, the word set in the domain of a linguistic variable is generated automatically by hedge algebras associated with a linguistic variable. Therefore, the cardinality of the generated word set is unlimited. However, in a specific period, a given number of words are selected to describe the numeric time series. The number of automatically generated words depends on the max specificity level k of words specified by human experts and the maximum word length in the word domain is equal to k .

In the proposed optimization model of LTS presented in [23], the word set used to describe the historical numeric time series data (so-called used word set) should be selected intuitively by human experts before the fuzziness parameter value optimization processes. Hence, the used word set depends on the human expert's intuition and may not be optimal. To select the optimal used word set to best describe historical numeric time series data from the linguistic variable domain, an optimization process should be executed. Besides, the FPDs should be optimized in parallel with the word-set selection to select the best numerical semantics of used words. Therefore, a co-optimization process for concurrently selecting the optimal used word set and FPDs is applied by utilizing PSO. Specifically, the co-optimization process includes inner and outer loops. The inner loop (inner PSO) optimizes the used word set while the outer loop (outer PSO) optimizes FPDs. Real encoding is used for both outer and inner loop. Specifically, in our experiments, only two linguistic hedges (*Little* and *Very*) are used to generate the word set of the linguistic variable domain, so the number of the optimized FPDs is only 2. They are the fuzziness measure of one of two generator words (e.g., $fm(c)$) and the fuzziness measure of one of two hedges (e.g., $\mu(Little)$). We have $fm(c) + fm(c^*) = 1$ and $\mu(Little) + \mu(Very) = 1$, so we can infer that $fm(c^*) = 1 - fm(c)$ and $\mu(Very) = 1 - \mu(Little)$. Each particle of the outer loop $Y_i^t = \{fm(c), \mu(Little)\}$, $i = 1, \dots, N$, in the swarm represents those two FPDs. For the inner loop, each particle corresponds to a solution represented as an array of real number $p_i = (w_1, \dots, w_{d_w}, w_j \in [0, 1])$. Each word x_l of the used word-set X_i is selected from the word-set W_{set} of $\mathcal{X}_L$ by zero-based index as follows:

$$X_i^t = \{x_l \in W_{set} \mid l = \lfloor w_j \times |W_{set}| \rfloor, 0 \leq l < |W_{set}|\} \quad (6)$$

where $\lfloor \cdot \rfloor$ denotes the integer part of a real number.

The details of all steps of the proposed co-optimization algorithm are as follows:

Algorithm 1. UWO (Gwmax, Wset, dw, δ) //Used word-set optimization procedure.

Input:

Parameters: G_{wmax} , W_{set} , d_w , δ ;

// G_{wmax} is the number of generations, W_{set} is the word-set of $\mathcal{X}_L$, d_w is the cardinality of the used word-set selected from W_{set} , δ is a set of FPVs.

Output: The optimal word-set $S_g^{G_{wmax}-1}$ and the associated best MSE value;

Begin

Randomly initialize a swarm $W_0 = \{Y_i^0; V_i^0 \mid i = 1, \dots, M, |Y_i^0| = d_w\}$;

// Y_i^t is a subset of index values in the interval $[0, |W_{set}|]$, where $|W_{set}|$ is the cardinality of W_{set} , $|Y_i^t|$ is the number of elements of Y_i^t .

Sort the elements of Y_i^0 ;

For each particle x_i in swarm **do begin**

Implement the LTS forecasting procedure from Step 2 to Step 6 based on δ and Y_i^0 ;

Evaluate the value of the MSE of x_i by equation (1);

Assign the personal best position S_i^0 of x_i to the current position;

End;

Assign the global best position S_g^0 to the best position in current swarm;

$t = 1$;

Repeat

For each particle x_i in swarm **do begin**

Compute new velocity V_i^t of x_i by equation (5);

Compute new position Y_i^t of x_i by equation (4);

Sort the elements of Y_i^t ;

Implement the LTS forecasting procedure from Step 2 to Step 6 based on δ and Y_i^t ;

Evaluate the value of the MSE of x_i by equation (1);

If Y_i^t is better than Y_i^{t-1} **then**

Update S_i^t of x_i based on the value of MSE;

End;

End;

Update S_g^t based on the values of MSE;

$t = t + 1$;

Until $t = G_{wmax}$;

Return the best position $S_g^{G_{wmax}-1}$ and its best MSE value;

End.

Algorithm 2. PSCO_FPVO //Fuzziness parameter value optimization.

Input: The designated time series dataset D ;

Parameters: N , G_{max} , G_{wmax} , the syntactical semantics of HAs, k_{max} , d_w ;

// G_{max} and G_{wmax} are the number of generations of outer and inner PSO, respectively; k_{max} is the maximum word length; d_w is the used word set's cardinality.

Output: The optimal FPVs $P_g^{G_{max}-1}$ and the best-used word-set W_{set}^* ;

Begin

Generate the word-set W_{set} of $\mathcal{X}_L$ with the maximum word length k_{max} utilizing HA ;

Randomly initialize a swarm $S_0 = \{X_i^0; V_i^0 \mid i = 1, \dots, N\}$, where $X_i^0 = \{fm(c^-), \mu(Little)\}$;

// X_i^t is a set of FPVs

//The best used word-set W_{set}^* and the best MSE value MSE^*

$(W_{set}^*, MSE^*) = (\emptyset, +\infty)$;

For each particle x_i **do begin**

$(W_{set}^*, MSE^*) = \text{UWO}(G_{wmax}, W_{set}, d_w, \delta=X_i^0)$; //call the inner PSO

$F_i^0 = MSE^*$; //Fitness value associated with particle x_i

Assign the personal best position P_i^0 of x_i to the current position;

End;

Assign the global best position P_g^0 the best position in current swarm;

$t = 1$;

Repeat

For each particle x_i **do begin**

Compute new velocity V_i^t of x_i by equation (5);

Compute new position X_i^t of x_i by equation (4);

$(W_{imp_set}, MSE) = \text{UWO}(G_{wmax}, W_{set}, d_w, \delta=X_i^t)$; //call the inner PSO

$F_i^t = MSE$; //Fitness value associated with particle i

If F_i^t is better than F_i^{t-1} **then begin**

Update the personal best P_i^t of x_i based on F_i^t 's value;

If MSE is better than MSE^* **then begin**

$(W_{set}^*, MSE^*) = (W_{imp_set}, MSE)$;

Update the global best position P_g^t ;

End;

End;

End;

$t = t + 1$;

Until $t = G_{max}$;

Return the best position $P_g^{G_{max}-1}$ and W_{set}^* ;

End.

As shown in the UWO and PSCO_FPVO algorithms, the PSCO_FPVO algorithm optimizes the FPVs and runs as an outer iteration of COLTS. Each particle of PSCO_FPVO represents a given FPV set $X_i^t = \{fm(c^-), \mu(Little)\}$, which is the input of UWO. In turn, UWO finds the optimally used word set from the word-set W_{set} of $\mathcal{X}_L$ based on the given FPV set as its input. Each particle of UWO represents a given used word set. The output of UWO includes the local best-used word set contained in $S_g^{G_{wmax}-1}$ corresponding to

the input FPV set and the associated local best MSE value. The output of PSCO_FPVO includes the best FPV set contained in $P_g^{G_{max}-1}$ and the global best-used word-set W_{set}^* , which is the best-found solution so far.

4. Experimental studies and discussions

The experiments aim at showing the necessity of optimizing the used linguistic word set used to describe numeric time series in parallel with optimizing FPVs and our proposed forecasting models outperform the compared ones. To show the efficiency of our proposed models, their experimental results on the three forecasting problems of the EUA, CAB, and GT are evaluated and compared with the ones proposed by Uslu et al. [6], Chen et al. [8], and Phong et al. [23]. The objective function of the optimization problem is the MSE. The values of MSE, RMSE, and MAPE values are used to evaluate the compared forecasting models.

In the experiments, our proposed forecasting models were implemented by C# programming language running on Microsoft Windows 10 64-bit with the hardware configuration of an Intel Core i5-8250U 1.6GHz CPU integrated with 8 GB of RAM. Each experiment has been attempted five times. Then, the smallest obtained MSE values are chosen as the forecasting accuracy.

For the inner iteration, the number of cycles is 100 and the number of particles is 30. For the outer iteration, the number of cycles is 30 and the number of particles is 20. For both inner and outer iterations, the Inertia weight ω is 0.4, both the self-cognitive factor c_1 and the social cognitive factor c_2 are 2.0.

4.1. Forecast the “enrollments of the University of Alabama”

To show the efficiency of the proposed LTS-FM described above, it is applied to forecast the EUA which is shown in Table 1. As in the counterparts, the UD of the EUA is $U = [13,000, 20,000]$, and the number of linguistics used to describe the numeric time series is 16. Instead of partitioning U into the intervals in the existing methods, it is transformed into linguistic words to form a LTS in our proposed forecasting methods. The linguistic word set is generated by the associated hedge algebras, so its cardinality is not limited. In fact, in a specific period, the number of words used to describe the numeric time series is limited. Therefore, PSO is applied to select the best-used word set to improve the accuracy of the forecasting model.

Table 1. Comparative simulation of enrollments’ forecast in case of 16 used words and the maximum word length of 5.

Year	Enrollments	CCO6 [5]	HPSO [7]	Uslu et al. [6]	Chen et al. [8]	Phong et al. [23]	COLTS3	COLTS4	COLTS5
1971	13055								
1972	13563	13714	13555	13650	13469	13515	13515	13562	13598
1973	13867	13714	13994	13650	13952	14001	14001	13759	13900
1974	14696	14880	14711	14836	14596	14800	14800	14722	14817
1975	15460	15467	15344	15332	15439	15509	15509	15412	15445
1976	15311	15172	15411	15447	15241	15509	15509	15464	15487
1977	15603	15467	15411	15447	15925	15509	15509	15464	15487
1978	15861	15861	15411	15447	15880	15752	15752	15798	15877
1979	16807	15831	16816	16746	16801	16693	16693	16799	16805
1980	16919	17106	17140	17075	17009	16949	16949	16975	16995
1981	16388	16380	16464	16380	16260	16779	16779	16431	16323
1982	15433	15464	15457	15504	15435	15553	15553	15412	15445

1983	15497	15172	15447	15431	15212	15509	15509	15464	15487
1984	15145	15172	15447	15077	15282	15132	15132	15286	15221
1985	15163	15467	15332	15297	15344	15132	15132	15312	15241
1986	15984	15467	16027	15848	15714	15752	15752	15824	15898
1987	16859	16831	16746	16835	16833	16693	16693	16833	16825
1988	18150	18055	18211	18145	18016	17888	17888	18193	18205
1989	18970	18998	19059	18880	18937	18911	18911	18833	18845
1990	19328	19300	19059	19418	19345	19439	19439	19246	19389
1991	19337	19149	19059	19260	19147	19307	19307	19143	19253
1992	18876	19149	19059	19031	19152	19043	19043	18936	18981
MSE		35324	31722	31684	23710	22403	22403	9755	6332
RMSE		187.95	178.11	178.00	153.98	149.68	149.68	98.77	79.57
MAPE		1.13%	0.84%	0.90%	0.73%	0.72%	0.72%	0.49%	0.40%

To have a significant comparison with the existing methods, we use in turn seven, fourteen, and sixteen linguistic words to describe U , and to show the significance of the word length, we limit the maximum word length, in turn, to 3, 4, and 5. The best MSE and MAPE values in accordance with the best word set selected by the inner iteration of the linguistic word optimization process and the best FPVs optimized by the outer iteration are shown in Table 2 and visualized by the comparison chart in Figure 2. We can see that with the same number of used words, the higher the maximum word length is, the better the MSE value we get. However, it is not absolutely true when slightly increasing the number of used words. In the case of the maximum word length of 3, the LTS-FMs with the number of used words of 14 is better than the one with the number of used words of 16 when compared by both MSE and MAPE values. In the case of the maximum word length of 4 and 5, the higher the number of used words is, the better the MSE value we get. Therefore, the best MSE is reached in the case of the number of used words of 16 and the maximum word length of 5.

Table 2. The MSE values of the first-order LTS-FMs with the different numbers of words and the different maximum word lengths.

The maximum word length	Evaluation method	Number of used words	Number of used words	Number of used words
		7	14	16
3	MSE	24111	19989	22415
	MAPE	0.80%	0.63%	0.72%
4	MSE	21284	10853	9758
	MAPE	0.74%	0.50%	0.49%
5	MSE	19795	9639	6332
	MAPE	0.72%	0.44%	0.40%

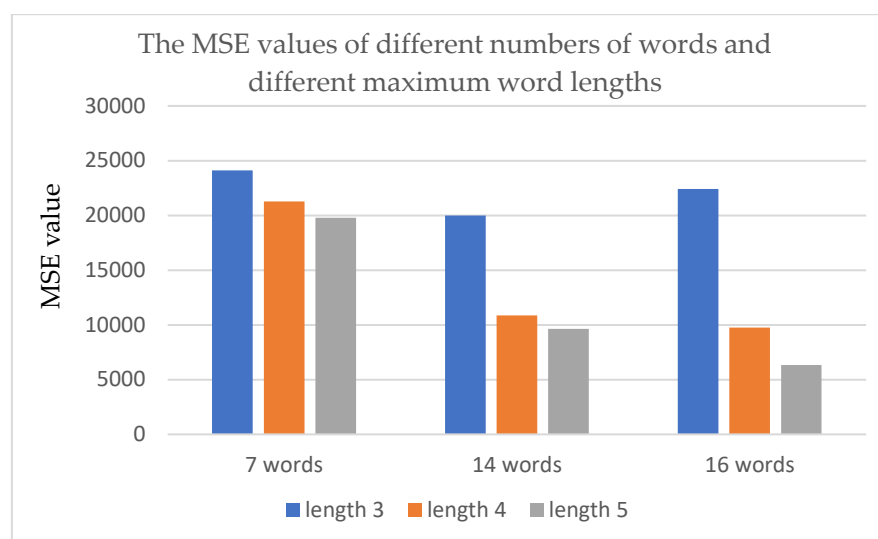


Figure 2. The MSE values with respect to the different numbers of words and the different maximum word lengths.

To show the efficiency of our proposed LTS-FMs with co-optimization PSO in comparison with the other forecasting models, the best experimental results of the proposed LTS-FMs with the maximum word length of 3, 4, and 5, denoted by COLTS3, COLTS4, and COLTS5, respectively, are compared with the state-of-the-art LTS-FM of Phong et al. [23] and the existing FTS-FMs such as CCO6 model of Chen and Chung applying genetic algorithm [5], HPSO [7] applying PSO algorithm, Uslu et al. [6] considering the number of iterations of a fuzzy logical relationship, and Chen et al. [8] applying PSO and new defuzzification technique. The experimental results and their comparative presentation are shown in Table 1. By analyzing the experimental results shown in Table 1, we can recognize that our proposed LTS-FM with the word length of 3 (COLTS3) has the same MSE and MAPE values as the LTS-FM proposed by Phong et al. in [23] and has better the MSE and the MAPE values of the rest FTS-FMs CCO6 [5], HPSO [7], Uslu et al. [6], and Chen et al. [8]. Meanwhile, both COLTS4 and COLTS5 have better MSE and MAPE values than those of the compared forecasting models. Therefore, we can conclude that the proposed LTS-FMs are better than the FTS-FMs and the COLTS5 is the best.

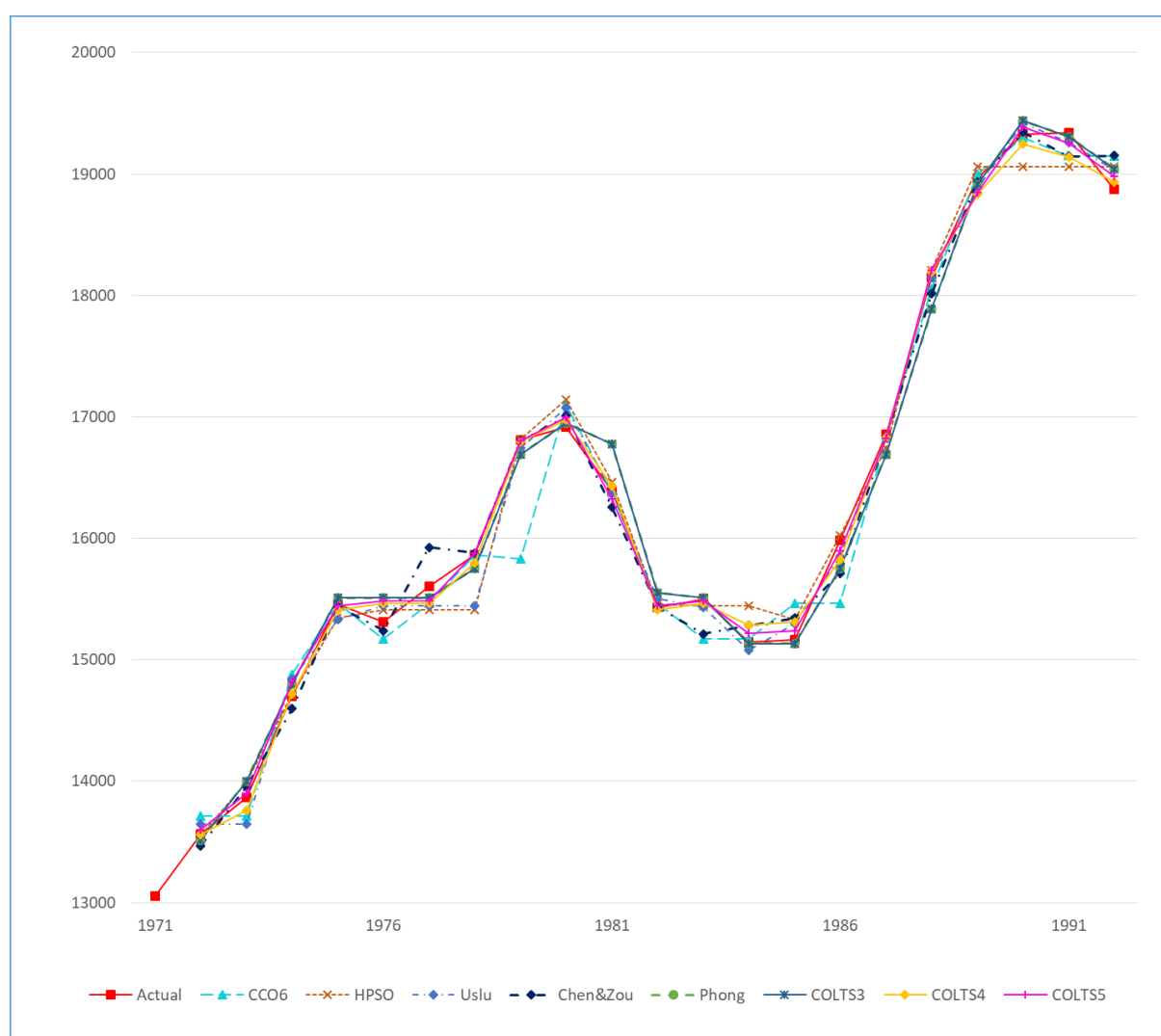


Figure 3. The comparison chart of the forecasted values of EUA of the compared forecasting models.

4.2. Forecast the “killed in car road accidents in Belgium”

The proposed LTS-FMs are also applied to the CAB’s problem to evaluate them again. The minimum and maximum values of the historical data of CAB observed from 1974 to 2004 are 953 and 1644, respectively. Thus, the UD is defined as $U = [900, 1700]$. The number of used words is set to 17 which is equal to the number of intervals in [6, 8].

Table 3. Comparative simulation of enrollments’ forecast in case of 16 used words and the maximum word length of 5.

Year	Actual data	Uslu et al. [6]	Chen et al. [8]	COLTS3	COLTS4	COLTS5
1974	1574					
1975	1460	1506	1451	1498	1495	1464
1976	1536	1453	1490	1500	1502	1515
1977	1597	1598	1622	1555	1592	1610
1978	1644	1584	1575	1593	1638	1623
1979	1572	1584	1593	1579	1568	1568
1980	1616	1506	1585	1593	1582	1610
1981	1564	1584	1582	1557	1560	1583

1982	1464	1506	1513	1485	1464	1462
1983	1479	1453	1494	1474	1471	1474
1984	1369	1375	1393	1423	1367	1381
1985	1308	1383	1336	1352	1333	1315
1986	1456	1454	1419	1450	1440	1462
1987	1390	1453	1485	1411	1417	1428
1988	1432	1383	1384	1419	1399	1391
1989	1488	1509	1459	1474	1479	1483
1990	1574	1598	1585	1534	1572	1580
1991	1471	1506	1451	1498	1500	1468
1992	1380	1375	1369	1411	1367	1381
1993	1346	1383	1361	1377	1333	1323
1994	1415	1383	1437	1400	1392	1381
1995	1228	1231	1217	1213	1276	1250
1996	1122	1135	1152	1136	1139	1156
1997	1150	1180	1172	1134	1122	1131
1998	1224	1245	1211	1225	1225	1187
1999	1173	1135	1147	1160	1158	1187
2000	1253	1245	1245	1249	1244	1246
2001	1288	1284	1280	1227	1281	1253
2002	1145	1143	1148	1158	1130	1163
2003	1035	970	1028	1058	1032	1032
2004	953	970	953	964	987	982
MSE		1731	1024	794	444	421
RMSE		41.61	32.00	28.18	21.07	20.52
MAPE		2.29%	1.77%	1.68%	1.253%	1.250%

It is easy to see from the comparison results in Table 3 that the MSE values of COLTS3, COLTS4, and COLTS5 are, in turn, 794, 444, and 421, much better than those of the compared forecasting methods of Uslu et al. [6] and Chen et al. [8], which are 1731 and 1024, respectively. When comparing by the MAPE values, we also see that all three of our LTS-FMs COLTS3, COLTS4, and COLTS5 have better MAPE values than those of Uslu et al. and Chen et al. presented in [6] and [8], respectively. Therefore, our proposed LTS-FMs outperform the FTS-FMs presented in [6, 8] in forecasting the problem of CAB. In addition, among our three LTS-FMs, COLTS5 is better than COLTS4 and, in turn, COLTS4 is better than COLTS3. Therefore, the statement “with the same number of used words, the higher maximum word length is, the better MSE value we get” presented above is also true for the CAB’s problem. The visual comparison chart of the compared forecasting models is shown in Figure 3.

4.3. Forecast the “spot gold in Turkey”

The proposed LTS-FMs are applied once again to a more complex forecasting problem of SGT. The minimum and maximum of the historical data of SGT observed from December 7th to November 10th are 30503 and 62450, respectively, so the UD is set to [30000, 63000]. The number of used words is 16 which is equal to the number of intervals in [6, 8].

Table 4. The forecasted values of the SGT of our proposed FTS-FMs are compared with those of the existing the forecasting models..

Date	Actual spot gold	Uslu et al. [6]	Chen et al. [8]	COLTS3	COLTS4	COLTS5
7-Dec	30503					
8-Jan	33132	32740.18	32341.38	32023.45	33145.28	33317.23
8-Feb	35201	34882.78	34479.36	34437.4	34368.77	34748.84
8-Mar	38529	37409.66	38605.47	38926.53	37751.7	38047.16
8-Apr	38300	39894.23	38203.34	39174.45	38551.01	38697.57
8-May	36142	37023.88	37406.67	36597.27	36770.38	37115.76
8-Jun	35837	37409.66	36749.36	35157.36	35971.07	36465.34
8-Jul	37074	37409.66	36452.85	37719.55	37151.04	37608.49
8-Aug	32955	32740.18	31805.51	33950.38	32537.23	32693.81
8-Sep	33277	34882.78	34335.42	34437.4	33745.95	33755.9
8-Oct	38295	37409.66	38120.71	37719.55	38545.47	38720.65
8-Nov	38677	37023.88	37402.31	37804.25	38551.01	38697.57
8-Dec	40724	39894.23	40726.33	40381.42	40353.8	40187.07
9-Jan	43985	43666.21	44515.67	43539.13	44427.03	43640.45
9-Feb	49931	49662.4	49800.77	47966.48	50125.47	49223.14
9-Mar	50823	51971.99	50962.66	51420.1	50985.28	51307.51
9-Apr	46167	45938.07	45869.8	46046.6	46786.96	46273.1
9-May	46716	46435.4	46548.24	46526.57	47031.66	46568.11
9-Jun	47337	46435.4	47067.02	46526.57	47031.66	46568.11
9-Jul	46088	46435.4	47653.83	46526.57	47031.66	46568.11
9-Aug	45839	46435.4	46473.59	46526.57	47031.66	46568.11
9-Sep	48053	46435.4	46238.3	47966.48	48255.15	48043.13
9-Oct	49592	49662.4	48330.41	50213.13	50501.27	50458.94
9-Nov	53693	51971.99	54338.06	52866.27	53035.95	53317.76
9-Dec	54553	54188.41	54509.96	54270.66	54472.2	54596.41
10-Jan	53022	54188.41	53663.01	53483.95	54472.2	53323.23
10-Feb	53613	54188.41	54183.79	54232.71	54472.2	53031.02
10-Mar	55031	54188.41	54471.07	54270.66	54472.2	54304.20
10-Apr	55181	54188.41	55887.68	55209.21	55316.72	54836.46
10-May	60300	60069.32	60030.78	60575.56	61184.06	60631.89
10-Jun	62100	60069.32	59888.46	61171.38	62123.95	60829.09
10-Jul	60500	59849.5	61610.89	61181.67	61184.06	60914.75
10-Aug	59200	60069.32	60079.84	59725.21	59495.78	59526.34
10-Sep	61250	60069.32	61520.74	61894.46	61184.06	61480.46
10-Oct	62450	62437.15	60797.52	62393.89	62123.95	61339.03
10-Nov	61600	59849.5	61945.8	61894.46	61184.06	61339.03
MSE		1,030,692	805,291	504,909	332,503	302,749
RMSE		1015.23	897.38	710.57	576.63	550.23
MAPE		1.80%	1.55%	1.38%	0.98%	1.00%

The experimental results of our proposed LTS-FMs shown in Table 4 are compared with those of Uslu et al. and Chen et al. presented in [6] and [8], respectively. We can see in Table 4 that when compared by both MSE and MAPE values, all three of our proposed LTS-FMs COLTS3, COLTS4, and COLTS5 outperform the FTS-FMs of Uslu et al. and Chen et al. presented in [6] and [8], respectively, in forecasting the problem of SGT. Besides, when compared by the MSE or RMSE values, COLTS5 is better than COLTS4, and COLTS3 is worse than COLTS4. However, when compared by MAPE value, COLTS5 is slightly worse than COLTS4. In general, once again, we can state that with the same number of used words, the higher the maximum word length is, the better the MSE value we get.

5. Conclusions

In this study, we have proposed a hybrid LTS-FM with a co-optimization procedure using PSO to determine applicable fuzziness parameter values and the best word set describing the observed data. When compared with the former studies, our proposed model gives the better forecasted accuracies. Based on this research, ones can continue studying to improve the forecasted accuracy of the forecasting model and apply to different forecasting application problems with various datasets. In addition, research on better data representation methods using LTS to improve the accuracy and the performance of the forecasting model is a good idea for future work.

Author Contributions: Conceptualisation, N.D.H. and P.D.P.; methodology, N.D.H. and P.D.P.; software, P.D.P. and M.V.L.; validation, M.V.L.; data curation, M.V.L. and N.D.H.; writing, P.D.P. and N.D.H.; visualization, M.V.L.; supervision, P.D.P. All authors have read and agreed to the published version of the manuscript.

Funding: This research is funded by Vietnam Ministry of Education and Training under grant number B2022-TTB-01.

Data Availability Statement: Not applicable.

Conflicts of Interest: The authors declare no conflict of interest.

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